

BIOL B215: BIOSTATISTICS WITH R

8. 1 Contingency Analysis

11/25/2025

Outline

- Quantifying and testing associations
 - Relative risk
 - Odds
 - Odds-ratio
- χ^2 contingency test
- Fisher's test

We Often Care About Associations

- Does eating chocolate decrease the chance you'll have a bad day?
- Does the home team have an advantage?
- Is the use of “bath salts” associated with cannibalism?   Probably not!*
- Does fertilizing tomatoes increase the chance they set fruit?
- Does a given drug affect the disease outcome in patients?

*Story: <https://www.smithsonianmag.com/science-nature/no-bath-salts-wont-turn-you-into-cannibal-180953418/>

Quantifying & Testing Associations

Between two categorical variables

Quantify:

- Relative risk
- Odds Ratio

Test:

- χ^2 contingency test
(or χ^2 test of independence)
- Fisher's exact test

Two kinds of chi-squared tests

Chi-square goodness-of-fit tests

- one variable (observed vs. expected)
- the chi-square distribution in tests where we compare observed frequencies for one variable in a sample to a theoretical expectation based on a distribution (e.g. the proportional model, the poisson model, etc)

Two kinds of chi-squared tests

Chi-square contingency (independence) tests (2x2)

- the chi-square statistic in tests that compare frequencies/counts for two categorical variables
- the χ^2 contingency test is a special application of the more general goodness-of-fit test for which the probability model being tested is the independence of variables

QUANTIFYING ASSOCIATIONS BETWEEN CATEGORICAL VARIABLES

- Relative risk
- Odds ratio

Relative Risk: Definition (1/)

- Relative risk is the probability of an undesired outcome in the treatment group divided by the probability of that outcome in the control group.

$$RR = \frac{p_1}{p_2}$$

$$\hat{RR} = \frac{\hat{p}_1}{\hat{p}_2}$$

$$\hat{p}_i = \frac{n_{\text{Bad outcomes in group } i}}{n_{\text{Individuals in group } i}}$$

- Probabilities that range from 0 to 1.
- RR ranges from 0 to ∞

Relative Risk: Definition (2/)

$$\hat{RR} = \frac{\hat{p}_1}{\hat{p}_2} \quad \hat{p}_i = \frac{n_{\text{Bad outcomes in group i}}}{n_{\text{Individuals in group i}}}$$

$\hat{RR} < 1$ Relative risk is lower in group 1

$\hat{RR} > 1$ Relative risk is higher in group 1

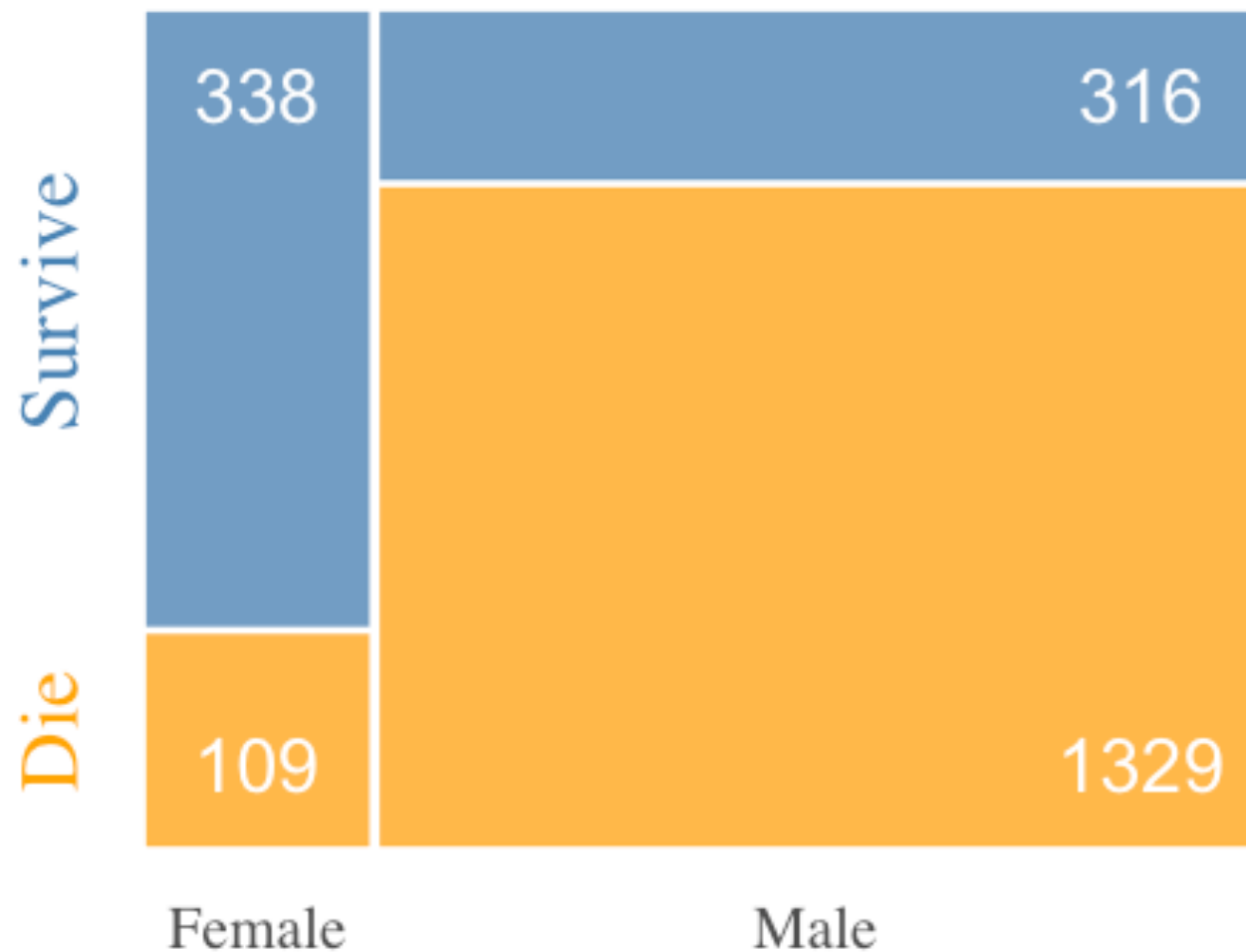
Reduction in relative risk is also of interest: $1 - \hat{RR}$

RELATIVE RISK VS ODDS-RATIO

TITANIC EXAMPLE!

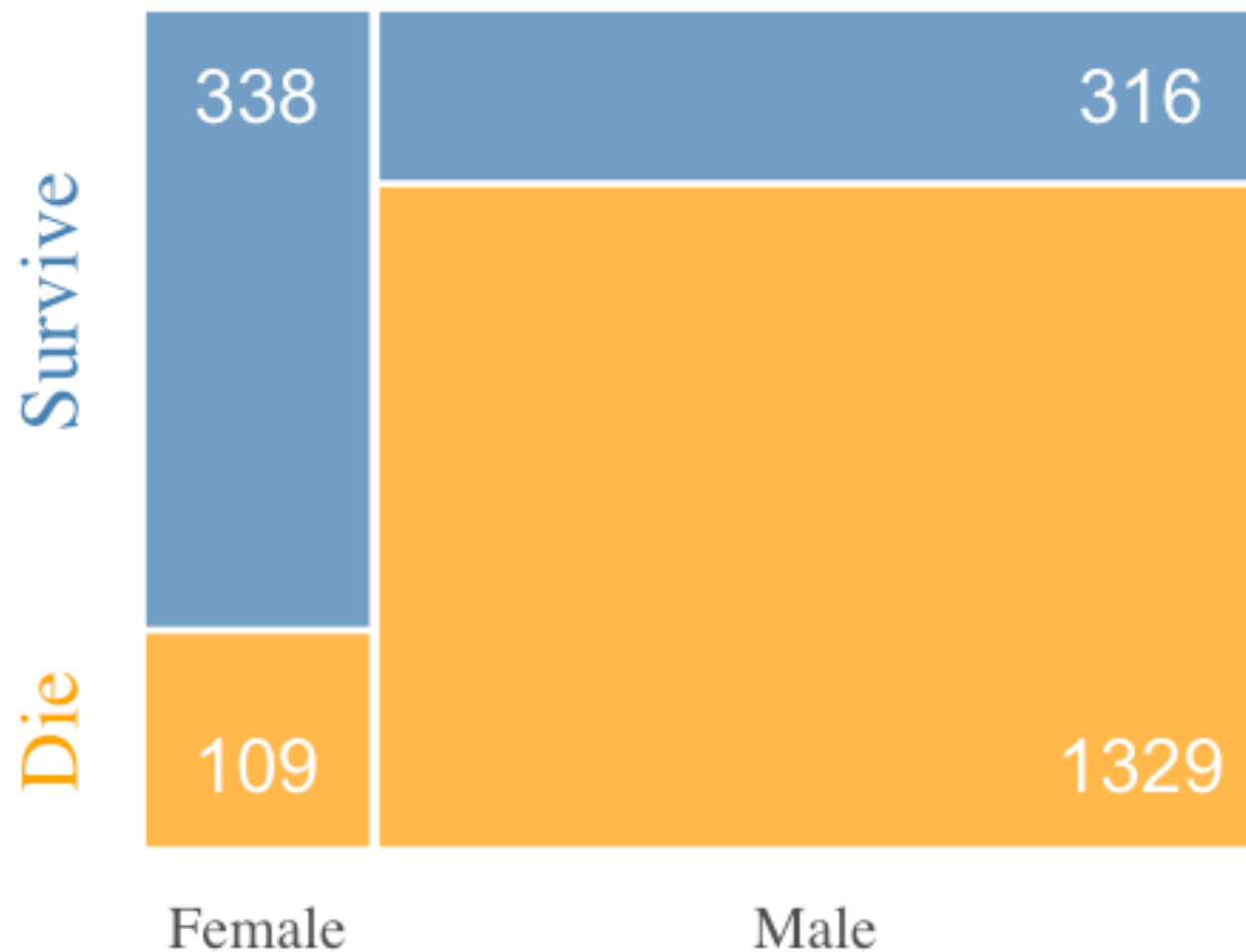


Outcome of interest: dying in the titanic



- What is being calculated? What is the focal outcome?
- What's the treatment group?
- What's the control group?
- How to interpret RR?
- How to interpret OR?

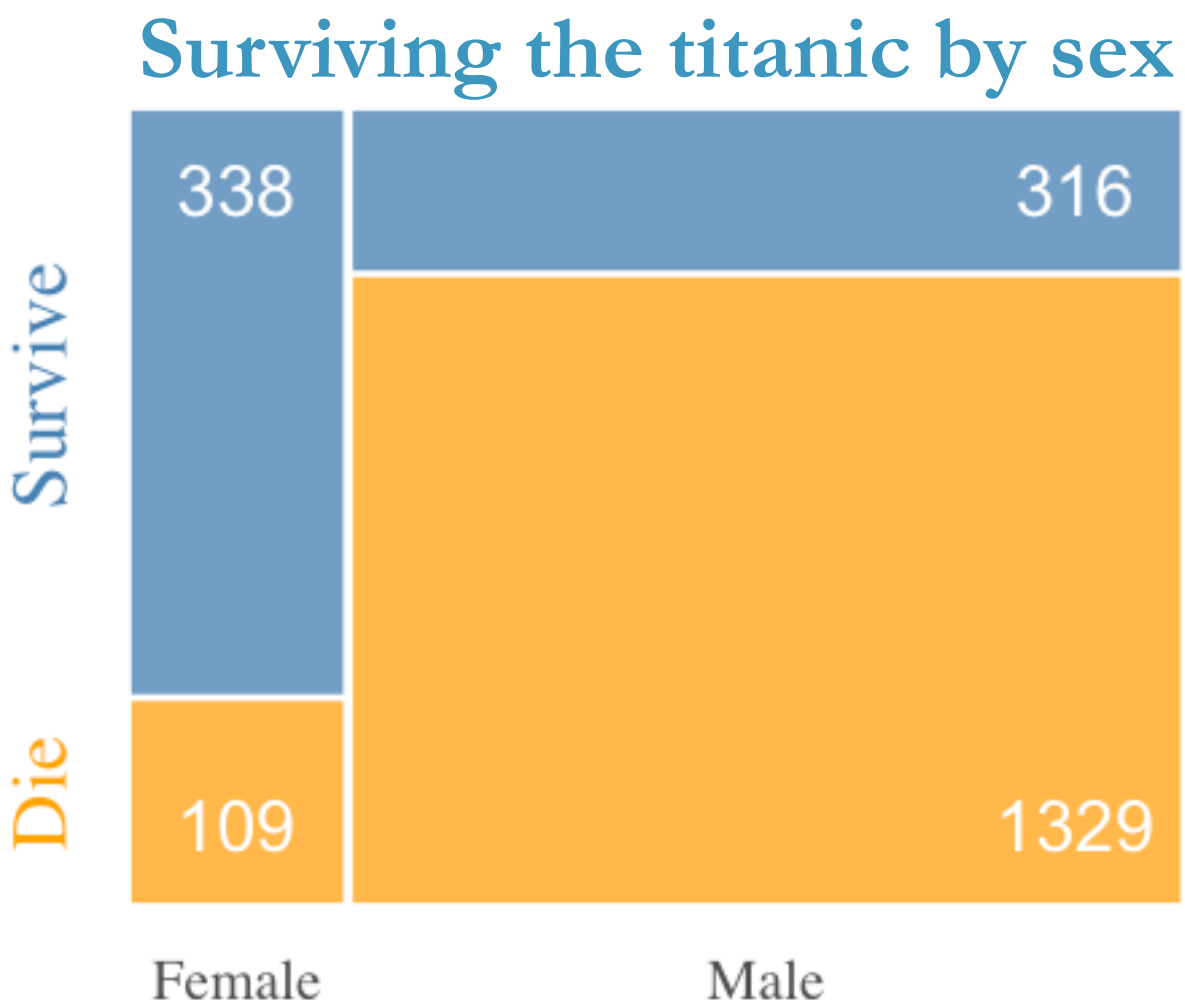
Outcome of interest: dying in the titanic



- What is being calculated? What is the focal outcome?
- What's the treatment group?
- What's the control group?

Titanic: Relative risk (1/)

- Relative Risk of dying in the titanic: female (treatment), male (control).



$$\hat{p}_{women} = \frac{death}{all\ women}$$

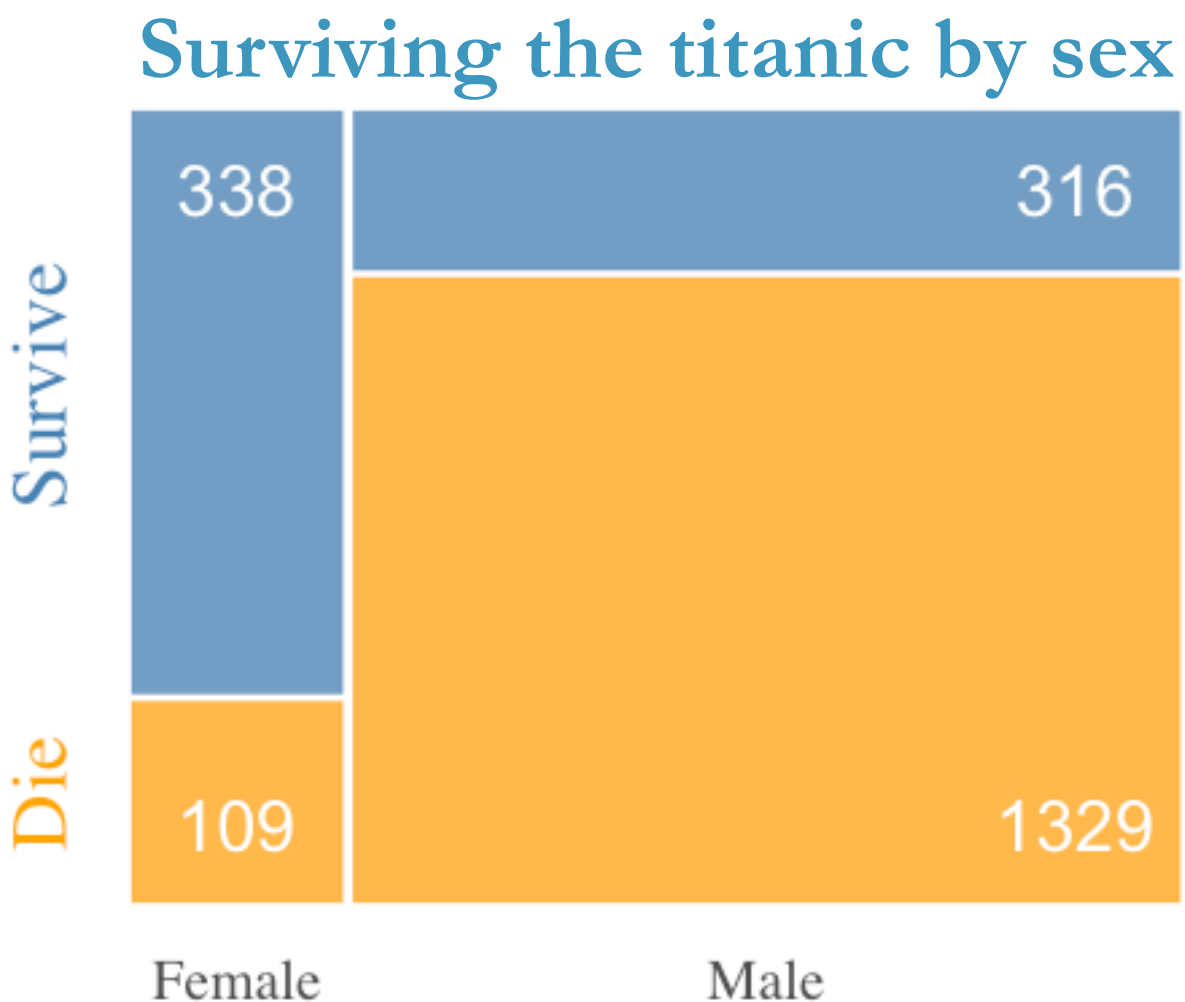
$$\hat{p}_{men} = \frac{death}{all\ men}$$

Women's relative risk of dying in the titanic compared to men

$$\hat{RR} = \frac{\hat{p}_{women}}{\hat{p}_{men}}$$

Titanic: Relative risk (2/)

- Relative Risk of dying in the titanic: female (treatment), male (control).



$$\hat{p}_{women} = \frac{109}{(109 + 138)} \approx 0.24$$

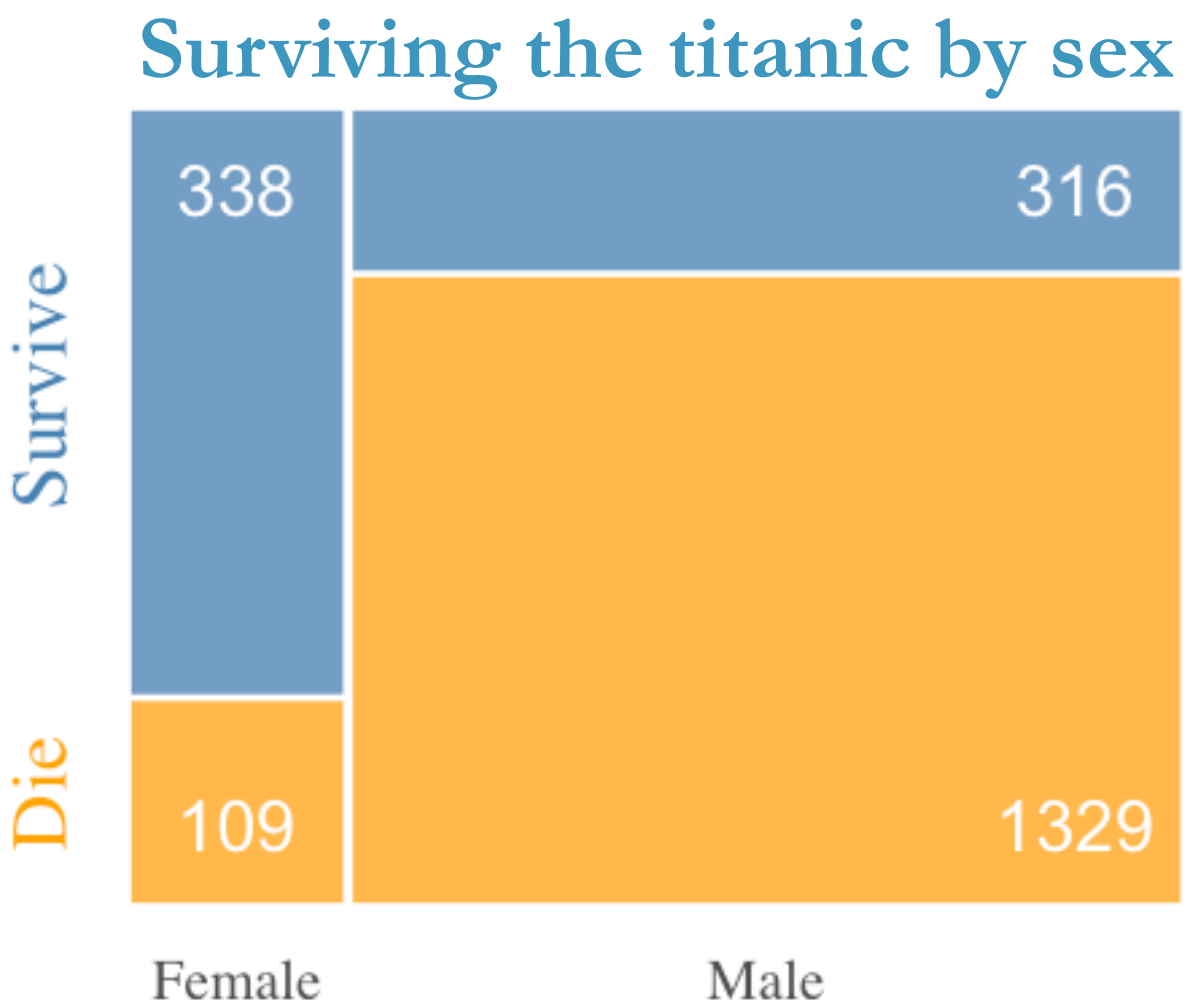
$$\hat{p}_{men} = \frac{1329}{(1329 + 316)} \approx 0.81$$

Women's
relative risk of
dying in the
titanic
compared to
men

$$\hat{RR} = \frac{0.24}{0.81} = 0.296$$

Titanic: Relative risk (3 /)

- Relative Risk of dying in the titanic: female (treatment), male (control).



$$\hat{p}_{women} = \frac{death}{all\ women}$$

$$\hat{p}_{men} = \frac{died}{all\ men}$$

Women's RR of dying

$$\hat{RR} = 0.296$$

Women's reduction in relative risk of dying compared to men

$$1 - \hat{RR} = 0.704$$

Relative Risk: Pros and Cons

Good

- Straightforward, intuitive interpretation
- Should be reported whenever probabilities (p_1, p_2) can be estimated without bias

Bad

- We often cannot study rare outcomes without some selection bias
- It has some non-desirable properties for statistical modeling.
- In those cases we might have to deal with the much less intuitive log Odds Ratio

Odds: Definition

- Probability of success divided by probability of failure
- “Success” refers to the focal outcome
- Note: Odds range from 0 to ∞ , unlike probabilities that range from 0 to 1.

Parameter

$$O = \frac{p}{1 - p}$$

$O > 1$ Success more likely

$O = 1$ Success and failure equally likely

Estimate

$$\hat{O} = \frac{\hat{p}}{1 - \hat{p}}$$

$O < 1$ Failure more likely

Odds Ratio

Contingency table: Cells contain counts

Refers to the focal outcome

The two categories identify the two groups whose probability of success is being compared.

	<i>Treatment</i>	<i>Control</i>
<i>Success</i>	a	b
<i>Failure</i>	c	d

This is a two-way contingency table

$$O_1 = \frac{P(\text{Success} | \text{Group}_1)}{1 - P(\text{Success} | \text{Group}_1)}$$

$$O_2 = \frac{P(\text{Success} | \text{Group}_2)}{1 - P(\text{Success} | \text{Group}_2)}$$

$$OR = \frac{O_1}{O_2}$$

$$\hat{OR} = \frac{\hat{O}_1}{\hat{O}_2}$$

Odds Ratio

Contingency table: Cells contain counts

Refers to the focal outcome

Success

Failure

Treatment

Control

a	b
c	d

The two categories identify the two groups whose probability of success is being compared.

This is a two-way contingency table

$$\hat{OR} = \frac{\hat{O}_1}{\hat{O}_2} = \frac{a/c}{b/d} = \frac{ad}{bc}$$

Titanic: odds-ratio example (1/)

- “Success” refers to the focal outcome: dying in the titanic

$$\hat{O} = \frac{p_{success}}{p_{failure}} = \frac{\hat{p}}{1 - \hat{p}}$$

$$\hat{O}_{women} = \frac{\hat{p}}{1 - \hat{p}} = \frac{0.24}{1 - 0.24} \approx 0.316$$

$$\hat{O}_{men} = \frac{\hat{p}}{1 - \hat{p}} = \frac{0.81}{1 - 0.81} \approx 4.263$$

Titanic: odds-ratio example (2/)

	<i>Women</i>	<i>Men</i>
<i>Death</i>	109	1329
<i>Survival</i>	338	316

$$\hat{OR} = \frac{\hat{O}_{women}}{\hat{O}_{men}} = ?$$

Titanic: odds-ratio example (3 /)

Route I

- “Success” refers to the focal outcome: dying in the titanic

	<i>Women</i>	<i>Men</i>
<i>Death</i>	109	1329
<i>Survival</i>	338	316

$$\hat{OR} = \frac{\hat{O}_{women}}{\hat{O}_{men}}$$

$$\hat{O}_{women} = \frac{\hat{p}}{1 - \hat{p}} = \frac{0.24}{1 - 0.24} \approx 0.316$$

$$\hat{O}_{men} = \frac{\hat{p}}{1 - \hat{p}} = \frac{0.81}{1 - 0.81} \approx 4.263$$

$$\hat{OR} = \frac{\hat{O}_{women}}{\hat{O}_{men}} = \frac{0.316}{4.263} \approx 0.08$$

Titanic: odds-ratio example (4/)

Route II

- “Success” refers to the focal outcome: dying in the titanic

	<i>Women</i>	<i>Men</i>
<i>Death</i>	109	1329
<i>Survival</i>	338	316

$$\hat{OR} = \frac{a/c}{b/d} = ?$$

$$\hat{OR} = \frac{a/c}{b/d} = \frac{109/338}{1329/316} = 0.0766782 \approx 0.08$$

Odds Ratio: Pros and Cons

Bad

- Not intuitive at all
- The odds ratio does not equal the relative risk.

Good

- The denominator drops out of the top and bottom – meaning this will work even if we don't have true absolute probabilities.
- This will be the case in studies of rare outcomes in which it makes sense to increase the number of these rare outcomes in our sample (as we do in case-control studies).

RELATIVE RISK VS ODDS-RATIO

Relative risk vs Odds ratio (1/)

Definition

RR

- Ratio of two **probabilities**
- Probability of undesired (focal) outcome in a treatment group divided by probability of the outcome in a control group

OR

- Ratio of two **odds**
- Odds of the focal outcome in the treatment group divided by the odds in the control group

Relative risk vs Odds ratio (2/)

Interpretation example

$$\hat{RR} = \frac{\hat{p}_{\text{women}}}{\hat{p}_{\text{men}}} \approx 0.296 \neq \hat{OR} = \frac{\hat{O}_{\text{women}}}{\hat{O}_{\text{men}}} \approx 0.08$$

- Relative probability of dying in the titanic is lower for women
- The value has intuitive meaning
- Odds of dying in the titanic are lower among women
- Value does not have intuitive meaning

PROBLEM: TOXOPLASMA AND CAR WRECKS

STEP-BY-STEP

Toxoplasma & Car Wrecks

(1/)

- Infectious disease caused by unicellular eukaryotic parasite
- One of the most common zoonoses throughout the world
- Usually transmitted to humans via cat feces or uncooked meat
- Dormant toxoplasmosis can cause prolonged reaction times of the muscles

Toxoplasmosis (*Toxoplasma gondii*)

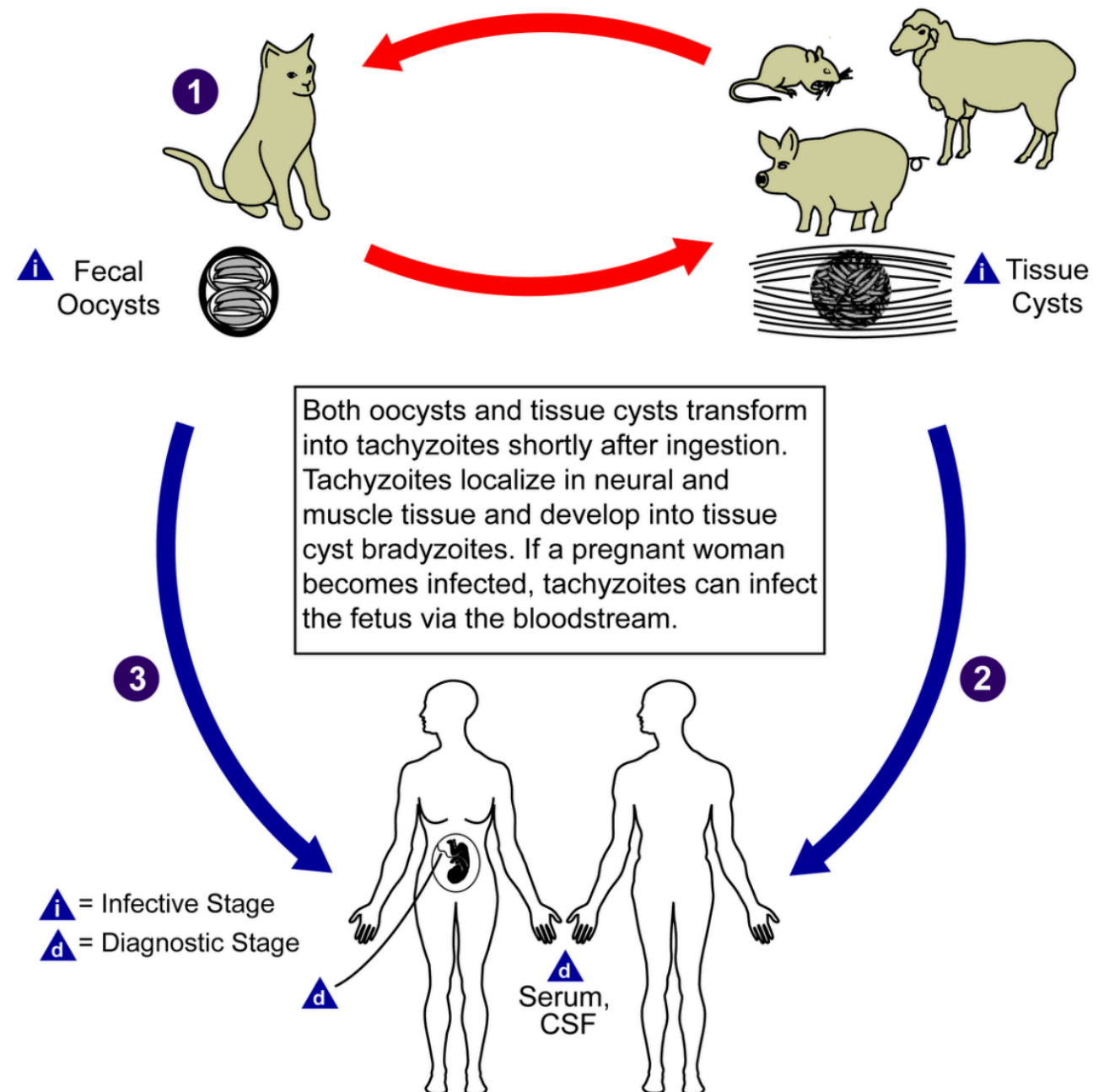


Image: [Wikipedia Commons](#)

Toxoplasma & Car Wrecks (2/)

- *Toxoplasma* tricks mice into taking risks so it can find its way into cats.
- Scientific question: Could it be associated with human risk taking?
- To test this, Yereli et al. (2006) compared *Toxoplasma* prevalence in drivers who caused car crashes to prevalence in those who hadn't.
- All drivers involved in accidents were tested for alcohol, cannabinoids, and opioids to control for confounding factors

Toxoplasma & Car Wrecks (3 /)

Does toxoplasma infection lead to car wreck?

- What we can look at: the proportion of people with and without toxoplasma among people in car wrecks and people not in car wrecks

	<i>Infected</i>	<i>Uninfected</i>
<i>Car wreck</i>	61	124
<i>No wreck</i>	16	169

$$\hat{OR} = \frac{\hat{O}_{\text{infected}}}{\hat{O}_{\text{uninfected}}}$$

$$\hat{OR} = \frac{61/16}{124/169} = \frac{3.81}{0.734} = 5.2$$

Toxoplasma & Car Wrecks (5 /)

Can we calculate the relative risk of being in a car wreck if infected? No

- We don't really know the proportion of people with toxoplasma who are involved in car accidents, so this is not appropriate

$\hat{p}_{\text{treatment}}$ = Proportion of people with toxoplasmosis who get involved in car wrecks

$\hat{p}_{\text{control}}$ = Proportion of people without toxoplasmosis who get involved in car wrecks

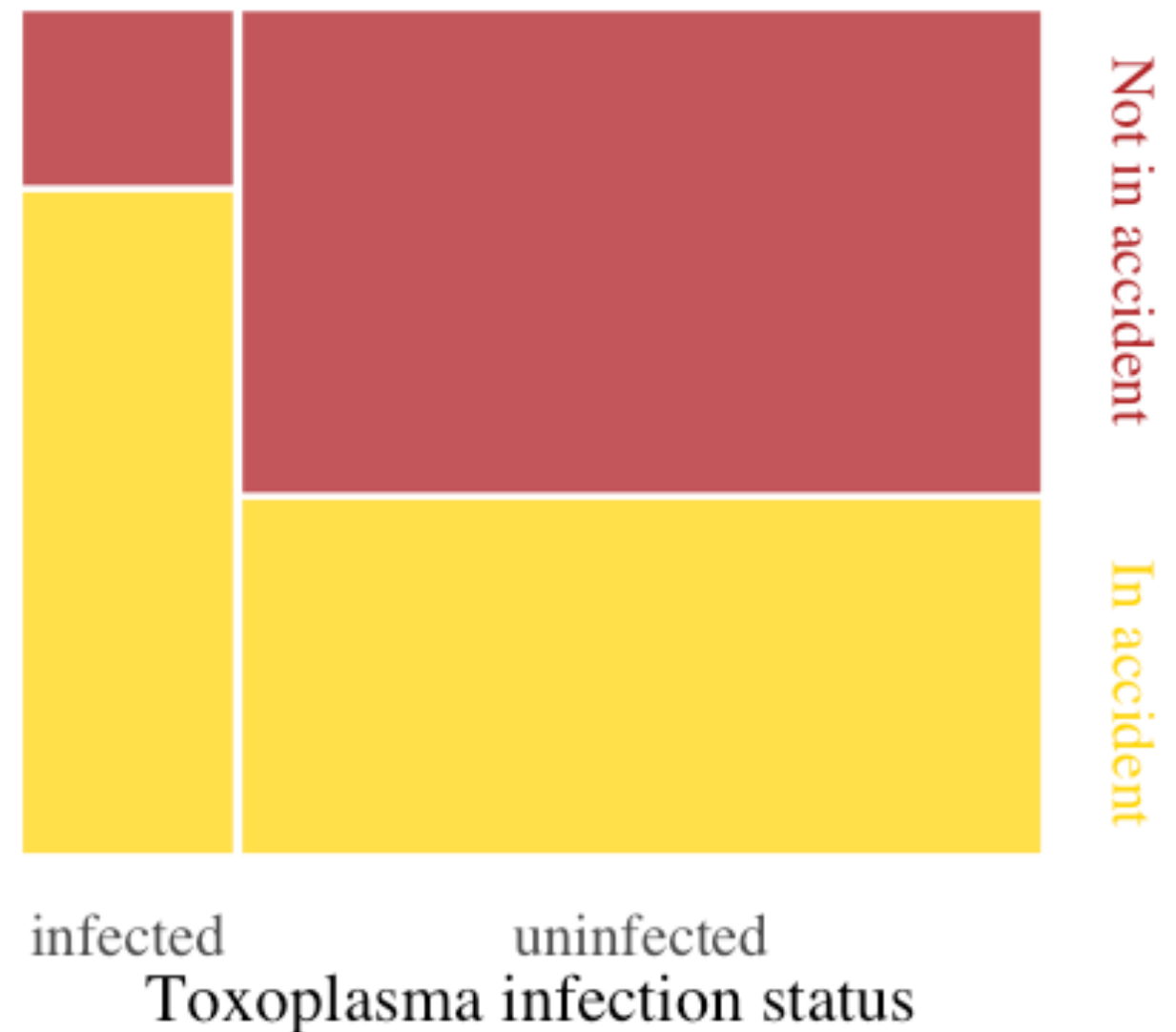
- What we do know is the proportion of people in car accidents that are infected versus proportion of infected people in a random sample of people not involved in car wreck

Toxoplasma & Car wrecks (11/)

Summary:

- Still, we learned that the odds of being in a car wreck is five times higher if you have toxoplasma than if you do not! (Odds ratio of car wreck among infected versus non-infected people)

More wrecks in toxoplasma cases



Toxoplasma & Car Wrecks: Summary

- We did not know the proportion of people with toxoplasma in car wrecks.
- Still, we learned that the odds of being in a car wreck is five times higher if you have toxoplasma than if you do not!
- We estimate a 5.2-fold increase in the chance of being in a car wreck for people with toxoplasma.
- We know that this estimate is uncertain.
- We can quantify this uncertainty!

Toxoplasma & Car wrecks

95% CI (1/)

- The standard error (SE) for the odds ratio cannot be directly computed, but the SE of the log odds can be!
- To find the 95% CI:
 1. Calculate $SE_{\ln OR}$
 2. Find 95% CI of $\ln(\hat{OR})$ as
 $[\ln(\hat{OR})] \pm 2 \times SE_{\ln OR}$
 3. Exponentiate to return to linear scale (if you like)

Toxoplasma & Car wrecks 95% CI (2/)

1. First, find the standard error of the log odds ratio:

$$SE_{\ln OR} = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

	<i>Infected</i>	<i>Uninfected</i>
<i>Car wreck</i>	61	24
<i>No wreck</i>	16	169

Toxoplasma & Car wrecks 95% CI (3/)

1. First, find the standard error of the log odds ratio:

$$SE_{\ln OR} = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

$$SE_{\ln OR} = \sqrt{\frac{1}{61} + \frac{1}{24} + \frac{1}{16} + \frac{1}{169}}$$

$$SE_{\ln OR} = 0.305$$

	<i>Infected</i>	<i>Uninfected</i>
<i>Car wreck</i>	61	24
<i>No wreck</i>	16	169

Toxoplasma & Car wrecks

95% CI (4/)

2. Find the 95% for $\ln OR$:

$$\begin{aligned}\ln[OR] &= \ln[OR] \pm 2 \times SE_{\ln OR} \\ &= \ln[5.2] \pm 2 \times 0.305 \\ &= 1.65 \pm 2 \times 0.305 \\ &\approx \{1.04, 2.36\}\end{aligned}$$

Toxoplasma & Car wrecks

95% CI (5 /)

3. Exponentiate to bring 95%CI of $\ln[OR]$ to linear scale

$$\ln[OR] = \approx \{1.04, 2.36\}$$

$$\log_b(a) = c; b^c = a$$

$$\{e^{1.04}, e^{2.36}\}$$

Euler's constant -
the base of the
natural logarithm

$$\{2.8, 10.6\}$$

OR 95% CI in
linear scale

Toxoplasma & Car wrecks 95% CI (6/)

$$2.8 < OR < 10.6$$

- The odds of getting in an accident are plausibly between 2.8 and 10.6 times higher for people with toxoplasma compared to those without.
- This 95% CI does not contain 1 (the null expectation)

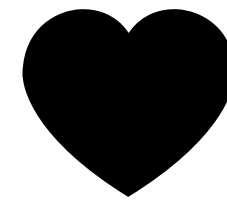
We reject the null hypothesis and conclude that car accidents & toxoplasma are associated*.

CAN WE INFER CAUSALITY FROM THIS?
DOES TOXOPLASMA INFECTION “CAUSE”
CAR ACCIDENTS?

NO! WE CANNOT KNOW THIS WITH THIS
INFORMATION ALONE!

WARNING: correlation does not necessarily equal causation. Assuming that it does can lead to very bad statistical practice!

95% CI in R with resampling (instead of this dreadful equation)



- There is another way: we can bootstrap!
- By bootstrapping, we approximate the sampling distribution by resampling outcomes with replacement and summarizing variability in this resampling to describe uncertainty.
- We will do this next lab (hopefully)
- We will use go over a case study on the drug Ribovaritin on SARS survival to apply these concepts

TESTING ASSOCIATIONS BETWEEN CATEGORICAL VARIABLES

- χ^2 contingency test (χ^2 test of independence)
- Fisher's exact test

χ^2 Contingency Test of Independence

Remember: The multiplication rule states that *if* A & B are independent:

$$P[A \& B] = P[A] \times P[B] - P[A \& B]$$

- This generates expectations of the null model.
- Use the χ^2 test to evaluate if deviations from this null expectation are surprising.

Calculating χ^2 & Degrees of Freedom

Remember: χ^2 is the sum of squared deviations between expectations and observations, *i and j* are rows and columns, respectively.

$$\sum_i \sum_j \frac{(\text{Observed}_{i,j} - \text{Expected}_{i,j})^2}{\text{Expected}_{i,j}}$$

- The degrees of freedom specify the additional number of values we need to know all the outcomes in our study.
- Degrees of Freedom for chi-square **contingency** test = $(n_{\text{rows}} - 1) \times (n_{\text{columns}} - 1)$

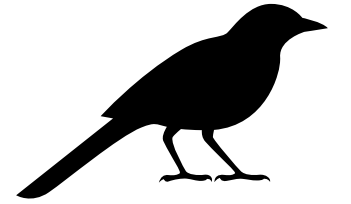
Example: Bird abandonment issues



- What's a bird to do when the rent is due and they can't feed their chicks?
- Kloskowski (2019) tested the idea that birds in poor habitats abandon their final egg.



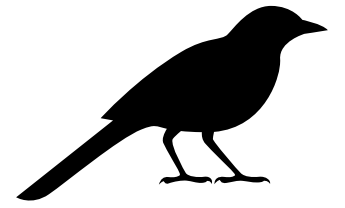
Bird abandonment Issues (1/)



H_0 : There is no association between habitat quality & the chance of abandoning the last egg

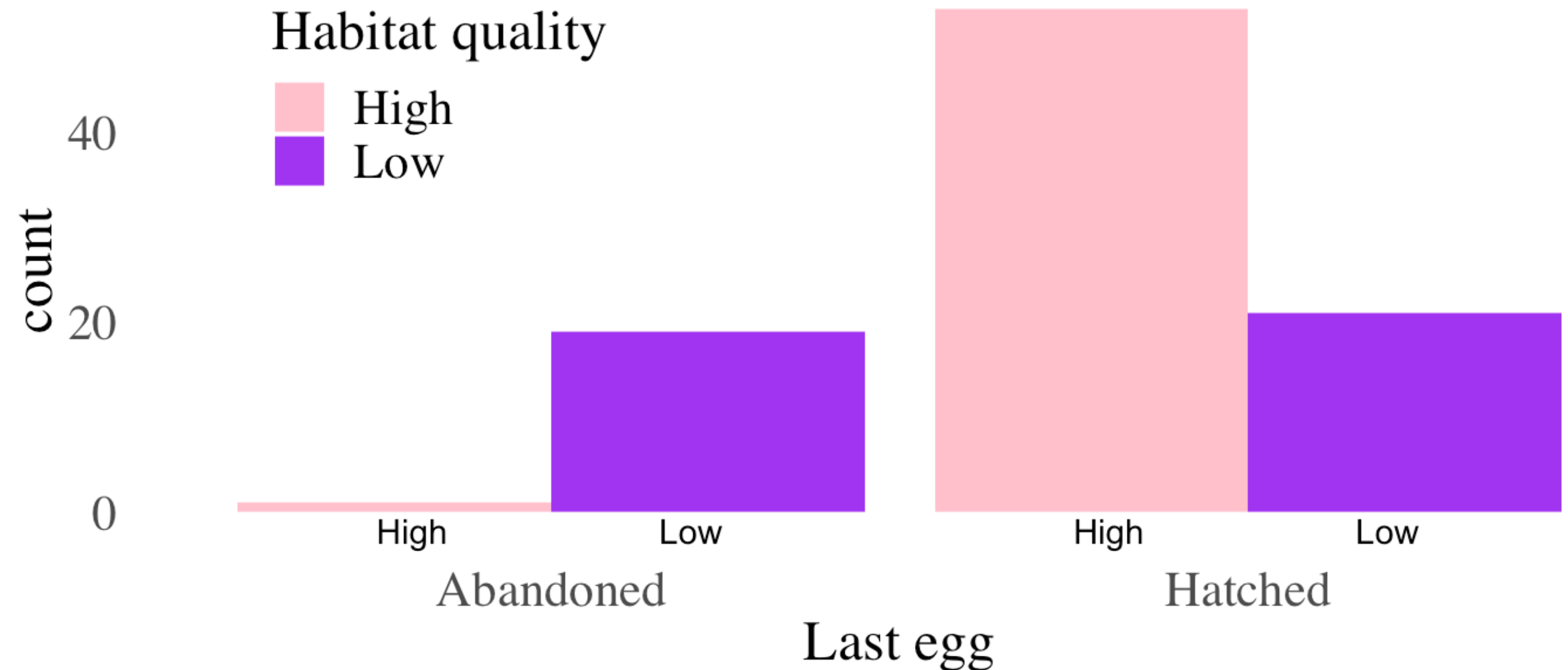
H_A : There is an association between habitat quality & the chance of abandoning the last egg

Bird abandonment Issues (2/)

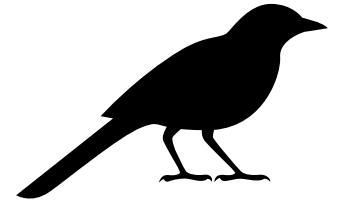


- Grouped barplot

Association between habitat quality & the proportion of last eggs abandoned.



Bird abandonment Issues (3 /)

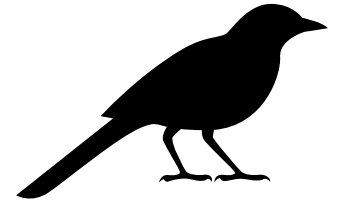


- Contingency table

Association between habitat quality & the proportion of last eggs abandoned.

	<i>Abandoned</i>	<i>Hatched</i>	<i>Total</i>
<i>Good quality habitat</i>	1	53	
<i>Good quality habitat</i>	19	21	
<i>Total</i>			

Bird abandonment Issues (4/)

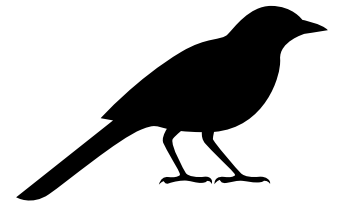


- Contingency table

Association between habitat quality & the proportion of last eggs abandoned.

	<i>Abandoned</i>	<i>Hatched</i>	<i>Total</i>
<i>Good quality habitat</i>	1	53	54
<i>Good quality habitat</i>	19	21	40
<i>Total</i>	20	74	94

Bird abandonment Issues (5/)

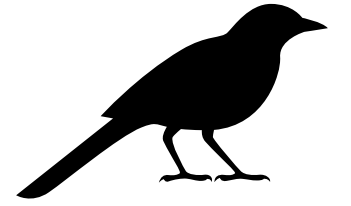


- Data analysis -friendly format.

<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$n_{Habitat}$	n_{Egg}
Good	Abandoned	1		
Good	Hatched	53		
Bad	Abandoned	19		
Bad	Hatched	21		

- What changed?

Bird abandonment Issues (6/)



- Data analysis -friendly format.

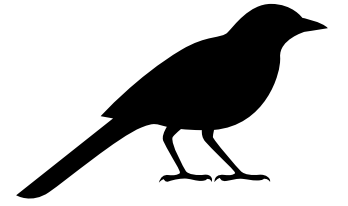
<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$n_{Habitat}$	n_{Egg}	$P_{Habitat}$	P_{Egg}
Good	Abandoned	1	54	20	0.57	0.21
Good	Hatched	53	54	74	0.57	0.79
Bad	Abandoned	19	40	20	0.43	0.21
Bad	Hatched	21	40	74	0.43	0.79

$$n_{total} = 94$$

$$P_{Habitat} = n_{Habitat} / n_{total}$$

$$P_{Habitat} = n_{egg} / n_{total}$$

Bird abandonment Issues (7 /)

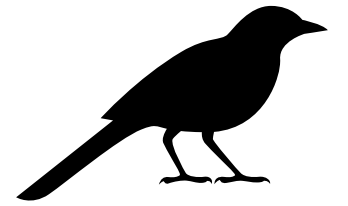


- Null expectations
- Find expectations under the null by multiplying proportions by each other and the sample size:

$$E[n(A, B) | n] = n \times P[A] \times P[B]$$

<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$P_{Habitat}$	P_{Egg}	n_{total}	<i>Expected</i>
Good	Abandoned	1	0.57	0.21	94	
Good	Hatched	53	0.57	0.79	94	
Bad	Abandoned	19	0.43	0.21	94	
Bad	Hatched	21	0.43	0.79	94	

Bird abandonment Issues (8/)

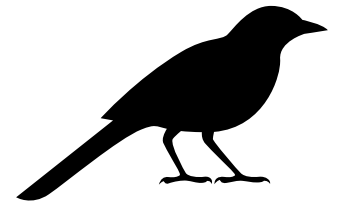


- Null expectations
- Find expectations under the null by multiplying proportions by each other and the sample size:

$$E[n(A, B) | n] = n \times P[A] \times P[B]$$

<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$P_{Habitat}$	P_{Egg}	n_{total}	<i>Expected</i>
Good	Abandoned	1	0.57	0.21	94	11.5
Good	Hatched	53	0.57	0.79	94	42.5
Bad	Abandoned	19	0.43	0.21	94	8.5
Bad	Hatched	21	0.43	0.79	94	31.5

Bird abandonment Issues (9/)



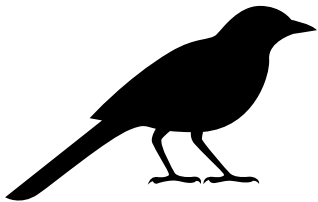
- Check chi-square test requirements
 1. Fewer than 20% of categories with Expected < 5
 2. No category with Expected ≤ 1

<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$P_{Habitat}$	P_{Egg}	n_{total}	<i>Expected</i>
Good	Abandoned	1	0.57	0.21	94	11.5
Good	Hatched	53	0.57	0.79	94	42.5
Bad	Abandoned	19	0.43	0.21	94	8.5
Bad	Hatched	21	0.43	0.79	94	31.5

We meet assumptions of the χ^2 test

Bird abandonment Issues

(10/)

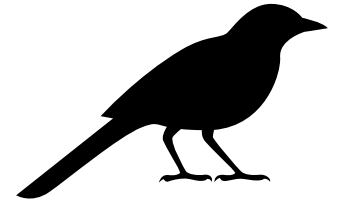


$$\sum \frac{(O - E)^2}{E}$$

<i>Habitat</i>	<i>Last Egg</i>	<i>Obs.</i>	$P_{Habitat}$	P_{Egg}	n_{total}	<i>Expected</i>
Good	Abandoned	1	0.57	0.21	94	11.5
Good	Hatched	53	0.57	0.79	94	42.5
Bad	Abandoned	19	0.43	0.21	94	8.5
Bad	Hatched	21	0.43	0.79	94	31.5

	<i>Good Abandoned</i>	<i>Good Hatched</i>	<i>Bad Abandoned</i>	<i>Bad Hatched</i>
Observed				
Expected				

Bird abandonment Issues (11/)



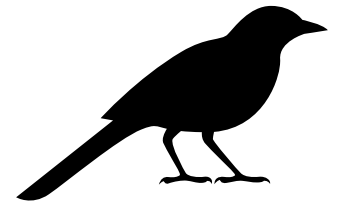
	<i>Good Abandoned</i>	<i>Good Hatched</i>	<i>Bad Abandoned</i>	<i>Bad Hatched</i>
Observed	1	53	19	21
Expected	11.5	42.5	8.5	31.5

$$\sum \frac{(O - E)^2}{E}$$

$$= \frac{(1 - 11.5)^2}{11.5} + \frac{(53 - 42.5)^2}{42.5} + \frac{(19 - 8.5)^2}{8.5} + \frac{(21 - 31.5)^2}{31.5}$$

$$= 28.5$$

Bird abandonment Issues (12/)



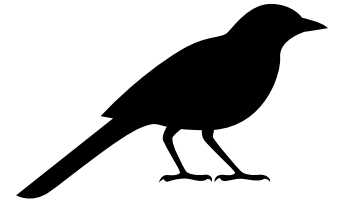
- Degrees of freedom

$$df = (n_{\text{columns}} - 1) \times (n_{\text{rows}} - 1)$$

$$df = (2 - 1) \times (2 - 1) = 1$$

- In each row or column of the contingency table, if we know $n-1$ of the values, we can infer the n -th value
- Therefore we subtract one df from each dimension

Bird abandonment Issues (13 /)

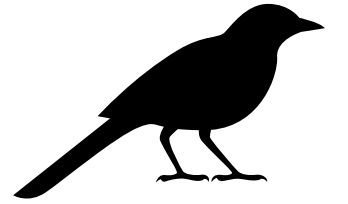


- Define alpha before. Then find critical value

Critical value for degrees of freedom and alpha level in columns

	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}
$df = 1$	2.7	6.6	10.8	15.1	19.5	23.9	28.4	32.8
$df = 2$	4.6	9.2	13.8	18.4	23.0	27.6	32.2	36.8

Bird abandonment Issues (14/)



- Define alpha before. Then find critical value

Critical value for degrees of freedom and alpha level in columns

	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}
$df = 1$	2.7	6.6	10.8	15.1	19.5	23.9	28.4	32.8
$df = 2$	4.6	9.2	13.8	18.4	23.0	27.6	32.2	36.8

- Significant at the $1e-7$ level
- Reject the H_0
- Conclude that bird abandon eggs in bad environments.

IN R

```
kableExtra::kable(birds.summary)
```

Habitat quality	Last egg	observed	p.habitat	p.egg	expected
High	Abandoned	10	57446810	21276611	489362
High	Hatched	530	57446810	78723442	510638
Low	Abandoned	190	42553190	212766	8.510638
Low	Hatched	210	42553190	78723431	489362

```
birds.chi<-birds.summary %>%  
  mutate(squared.dev = (observed - expected)^2/expected) %>%  
  summarise(chi2      = sum(squared.dev)) %>%  
  pull()  
birds.chi
```

```
## [1] 28.58684  $\chi^2$  without rounding anywhere
```

```
pchisq(q = birds.chi, df = 1, lower.tail = FALSE)
```

```
## [1] 8.958879e-08
```

IN R

```
kableExtra::kable(birds.summary)
```

Habitat quality	Last egg	observed	p.habitat	p.egg	expected
High	Abandoned	10	574468	10.212766	11.489362
High	Hatched	530	574468	10.787234	42.510638
Low	Abandoned	190	425531	90.212766	8.510638
Low	Hatched	210	425531	90.787234	31.489362

```
birds.mat<-matrix(birds.summary$observed, nrow=2)  
chisq.test(birds.mat,correct=F)
```

```
##  
## Pearson's Chi-squared test  
##  
## data: birds.mat  
## X-squared = 28.587, df = 1, p-value = 8.959e-08
```

χ^2
**without
rounding
anywhere**

One Tailed Test?

- We only looked at one tail of the χ^2 distribution.
- **This does not mean that we only notice an association in one direction.**
- Both ways of being exceptional (an association between abandonment and poor environments OR an vice versa) yield a strong discrepancy between observations and expectations, and can generate a significant association.
- To get a sense for the effect size we should conduct estimation

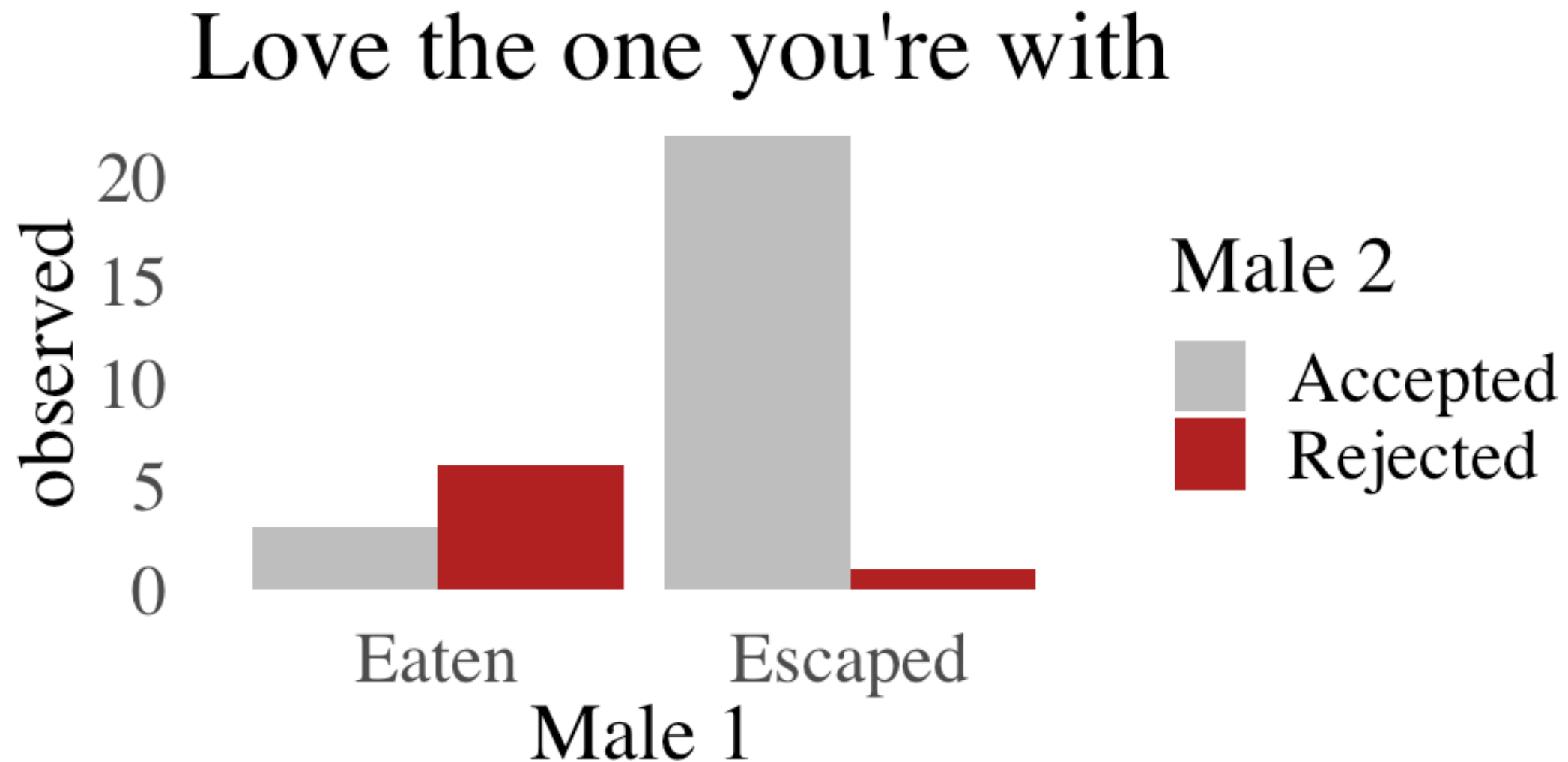
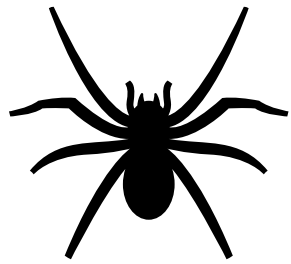
Fisher's test Example: Nuptial Gifts



- Sometimes female *Latrodectus hasselti* (redback spiders) eat their mates. Is there anything in it for the males?
- Maydianne Andrade tested the idea that eating a male might prevent the female from re-mating with a second male.

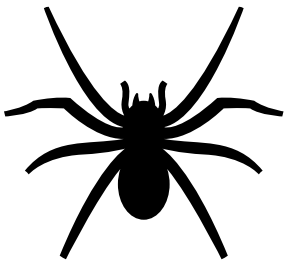


Nuptial Gift (1/)



- Eaten/escaped refers to male 1
- Accepted/rejected refers to male 2

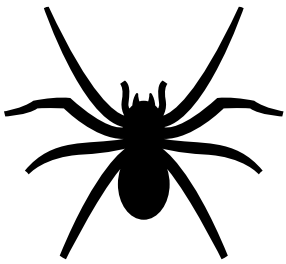
Nuptial Gift (2/)



- Data analysis -friendly format.

1 st Male	2 nd Male	Observed	Expected
Eaten	Accepted	3	
Eaten	Rejected	6	
Escaped	Accepted	22	
Escaped	Rejected	1	

Nuptial Gift (2/)



Can we do a χ^2 independence test?

- Fewer than 20% of categories with Expected < 5
- No category with Expected ≤ 1

Male 2

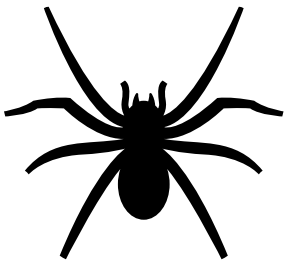
Male 2

Eaten

Escaped

Accepted	Rejected
3	6
22	1

Nuptial Gift (3 /)



Can we do a χ^2 independence test?

- Fewer than 20% of categories with Expected < 5
- No category with Expected ≤ 1

We do not
meet the
assumptions

Male 2

Eaten

Escaped

Male 2

Accepted	Rejected
3	6
22	1

Chi-square independence test vs Fisher's exact test (1/3)

- Chi-Square Test of Independence:
 - more traditional hypothesis test
 - uses a test statistic and its sampling distribution to calculate the p-value.
 - The chi-square sampling distribution only *approximates* the correct distribution, providing better p-values as the cell values in the table increase.
 - Consequently, chi-square p-values are invalid when you have small cell counts.

Chi-square independence test vs Fisher's exact test (2/3)

- Fisher's exact test:
 - doesn't use the chi-square statistic and sampling distribution.
 - It calculates the number of **all possible contingency tables with the same row and column totals** (i.e., **marginal distributions**) as the observed table.
 - Then it calculates the probability for the p-value by finding the proportion of possible tables that are more extreme than the observed table.
 - Theoretically appropriate for all sample sizes. However, the number of possible tables grows at an exponential rate and soon becomes unwieldy.
Hence, statisticians use this test for smaller sample sizes.⁷¹

Chi-square independence test vs Fisher's exact test (3/3)

- Both have the same null hypothesis: the outcome is independent from the groups.
- Chi-square is generally best for larger samples and Fisher's is better for smaller samples.
- Both can be extended to cases beyond 2x2 with 2 categorical variables, such as 3x2, 2x5, etc
- When to use Fisher's exact test:
 - $> 20\%$ of categories with Expected values < 5
 - Any expected value < 1
 - The column or row marginal values are extremely uneven.

G-tests (1/2)

- Another alternative to χ^2 contingency test for testing associations between categorical variables
- Makes no assumptions about cell sizes.
- Hypotheses are the same as in the independence test
- It's been around since 1981 but for some reason has not replaced Pearson's χ^2 2x2 test, even though it's better
- For samples of a reasonable size, the G-test and the chi-squared test will lead to the same conclusions.
- When to use:
 - Honestly, it's always better than χ^2 but
 - When χ^2 requirements are not met and counts are very high (because that makes Fisher's test very time-consuming)

G-tests (2/2)

- The commonly used **chi-squared tests** for goodness of fit to a distribution and for independence in **contingency tables** are in fact approximations of the **log-likelihood ratio** on which the G-tests are based

$$G = 2 \sum_i O_i \ln \left(\frac{O_i}{E_i} \right)$$

- When O and E are similar, $G \approx \chi^2$ with the same degrees of freedom calculation.
- When this difference is large, however, the approximation begins to break down. Here, the effects of outliers in data will be more pronounced, and this explains the why χ^2 tests fail in situations with little data.

Final thoughts

- The relative risk and odds ratio quantify the association between categorical variables.
- χ^2 quantifies the evidence for a deviation from expectations.
- An identical estimated relative risk of 2:1 would have a way bigger χ^2 value if it came from two-thousand observations than if it came from twenty observations.
- χ^2 does not capture the effect size in the same way as RR or OR
- The RR and OR values are very similar when the focal outcome is rare